AIGP Module 2 Advanced Practice Test

AI Impacts on People and Responsible AI Principles

30 Medium to Hard Questions (Exam Simulation)

Question 1

An AI recruiting system trained on 10 years of historical hiring data consistently ranks candidates from certain universities higher than others. Investigation reveals that the company historically hired more from these universities, creating a feedback loop. What type of bias does this PRIMARILY represent?

A. Implicit bias
B. Sampling bias
C. Temporal bias
D. Overfitting to training data

Correct Answer: B

Explanation:

- **Why B is correct:** Sampling bias occurs when "Data gets skewed toward a subset of a group and therefore may favor that subset of a larger group." The historical data over-represents candidates from certain universities because past hiring patterns created a skewed dataset. The training data doesn't represent the full population of qualified candidates—it reflects the company's historical preferences, creating sampling bias that perpetuates through the AI system.
- **Why A is incorrect:** Implicit bias refers to "Discrimination or prejudice toward a particular group or individual" by the developers themselves. While the outcome is discriminatory, the root cause is skewed training data (sampling), not developer prejudice.
- **Why C is incorrect:** Temporal bias occurs when "A model is trained and functions properly at the time, but may not work well at a future point." The issue here isn't that the data became outdated over time—it's that the data was never representative of the full candidate pool.
- **Why D is incorrect:** Overfitting means "A model works for the training data, but does not work for new data because it is so fitted to the training data." The model is actually working as designed on new data—the problem is the training data itself was biased, not that the model overfitted to training patterns.

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Question 2

According to the OECD AI Principles, which principle MOST directly addresses the requirement that AI systems should respect human rights, fundamental freedoms, and democratic values including fairness and privacy?

A. Inclusive growth, sustainable development, and well-being
B. Human rights and democratic values
C. Transparency and explainability
D. Robustness, security, and safety

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly lists OECD AI Principle #2 as "Human rights and democratic values (including fairness and privacy)." This principle directly addresses respect for human rights, fundamental freedoms, democratic values, fairness, and privacy as core requirements for AI systems.
- **Why A is incorrect:** While "Inclusive growth, sustainable development, and well-being" (Principle #1) addresses positive societal outcomes, it focuses on economic and social benefits rather than specifically on human rights, democratic values, and fundamental freedoms.
- **Why C is incorrect:** "Transparency and explainability" (Principle #3) addresses the need for understandable AI systems but doesn't directly speak to human rights, democratic values, or fundamental freedoms as its primary focus.
- **Why D is incorrect:** "Robustness, security, and safety" (Principle #4) focuses on technical reliability and protection from harm, not specifically on human rights and democratic values.

Question 3

A healthcare AI system trained in 2018 on patient data and treatment protocols now shows declining diagnostic accuracy in 2024. Medical practices, treatment guidelines, and disease patterns have evolved significantly. Which type of bias is this scenario MOST clearly demonstrating?

A. Sampling bias
B. Implicit bias
C. Temporal bias
D. Edge case bias

Correct Answer: C

Explanation:

- **Why C is correct:** This is a textbook example of temporal bias: "A model is trained and functions properly at the time, but may not work well at a future point, requiring new ways to address the data." The system worked well in 2018 with 2018 medical

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knowledge and disease patterns, but medical practice evolves. Treatment protocols change, new conditions emerge, and disease patterns shift, making historical training data less relevant over time.

- **Why A is incorrect:** Sampling bias relates to skewed representation in training data toward a subset of the population. The issue here isn't that certain groups were underrepresented—it's that the entire dataset has become outdated due to the passage of time and medical advancement.
- **Why B is incorrect:** Implicit bias refers to prejudice by developers toward particular groups. The accuracy decline affects diagnostic performance generally, not specific demographic groups, and results from outdated data rather than developer prejudice.
- **Why D is incorrect:** Edge cases refer to data points outside normal training boundaries or unusual situations. The problem affects general performance as medical knowledge evolved, not just unusual or outlier cases.

Question 4

An organization is evaluating privacy-enhancing technologies (PETs) for an AI system that must analyze sensitive financial data from multiple banks without centralizing the raw data. Which PET would be MOST appropriate for this scenario?

A. Differential privacy
B. Federated learning
C. Homomorphic encryption
D. De-identification

Correct Answer: B

Explanation:

- **Why B is correct:** Federated learning is specifically designed for "training models across multiple data sets without centralizing the raw data." Each bank can keep its sensitive data locally while contributing to a shared model. The training process sends model updates (not raw data) to a central coordinator, enabling collaborative learning without data centralization—exactly what this scenario requires.
- **Why A is incorrect:** Differential privacy "adds noise to protect individual data points" but doesn't solve the problem of analyzing data across multiple institutions without centralization. You could use differential privacy within federated learning, but it alone doesn't address the distributed data challenge.
- **Why C is incorrect:** Homomorphic encryption "enables computation on encrypted data," which protects data during processing but doesn't inherently prevent centralization. You'd still need to gather encrypted data centrally, which creates different governance and security challenges.
- **Why D is incorrect:** De-identification removes identifying information but doesn't address the core requirement of analyzing data across multiple institutions without centralizing it. De-identified data would still need to be collected centrally, and re-identification risks remain.

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Question 5

A facial recognition system achieves 98% accuracy overall but only 85% accuracy for individuals with darker skin tones and 87% for women. Which responsible AI principle is MOST directly violated, and what is the appropriate governance response?

A. Transparency; publish accuracy rates publicly
B. Fairness; conduct disaggregated testing, implement bias mitigation, and consider whether deployment is appropriate
C. Accountability; assign a responsible individual
D. Privacy; implement data minimization

Correct Answer: B

Explanation:

- **Why B is correct:** This scenario demonstrates differential performance across demographic groups—a fairness violation. The training notes "Facial recognition in healthcare, training data biases can result in misdiagnosis; NIST and London police facial recognition studies show unreliability for females and those with darker skin tones." Fairness requires that AI systems don't discriminate or perform worse for protected groups. The appropriate response includes: disaggregated testing (which was done, revealing the disparity), bias mitigation techniques, improved training data diversity, and potentially reconsidering deployment if fairness can't be ensured. The training emphasizes "protection from unfair bias" as a core responsible AI principle.
- **Why A is incorrect:** While transparency is important and publishing accuracy rates would be good practice, it doesn't address the underlying fairness problem. Transparency without remediation means openly deploying a discriminatory system.
- **Why C is incorrect:** Accountability is important for who's responsible for addressing the issue, but it doesn't specify what the problem is or how to fix it. The primary violation is fairness, though accountability mechanisms would be part of the governance response.
- **Why D is incorrect:** Privacy and data minimization aren't the primary concerns here. The issue is discriminatory performance, not excessive data collection or privacy violation.

Question 6

According to Fair Information Practices (FIPs), which principle requires that personal data should only be collected for specified purposes and not used for other purposes without consent?

A. Data minimization/collection limitation
B. Use limitation
C. Purpose specification
D. Safeguards/security

Correct Answer: B

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Explanation:

- **Why B is correct:** "Use limitation" specifically addresses that data should only be used for the purposes for which it was collected, and different uses require consent or authorization. The training lists "Use limitation" as a distinct FIP principle. This prevents function creep where data collected for one purpose gets repurposed without subject knowledge or consent.
- **Why A is incorrect:** "Data minimization/collection limitation" addresses collecting only what's necessary and not collecting excessive data. While related, it focuses on the amount and necessity of collection, not restrictions on subsequent use.
- **Why C is incorrect:** "Purpose specification" requires that purposes for data collection be specified at collection time. While related to use limitation, it's about stating purposes upfront, not restricting subsequent uses to those purposes.
- **Why D is incorrect:** "Safeguards/security" addresses protecting data from unauthorized access, loss, or misuse through security measures, not restricting uses to specified purposes.

Question 7

An AI governance professional discovers that a predictive policing system is creating a feedback loop: the system predicts crime in certain neighborhoods, police increase patrols there, more arrests occur, and the system learns to predict even more crime in those areas. What type of harm does this PRIMARILY represent?

A. Individual harm - economic opportunity
B. Group harm - deepening racial/socio-economic divides
C. Organizational harm - reputational risk
D. Environmental harm - resource consumption

Correct Answer: B

Explanation:

- **Why B is correct:** The training specifically identifies "Group harms" including "Deepening of racial/socio-economic divides." Predictive policing feedback loops disproportionately impact communities, often minority and lower-income neighborhoods, creating systemic discrimination. This isn't just individual harm—it affects entire communities and reinforces existing social inequalities. The training notes "Civil rights (e.g., tracking/profiling of protestors)" and "Mass surveillance risks" as group-level harms that this scenario exemplifies.
- **Why A is incorrect:** While individuals in affected neighborhoods experience harm, the primary concern is the systemic, community-wide impact that deepens social divides. The harm operates at the group/community level, not just individual economic opportunity.
- **Why C is incorrect:** Organizational reputational harm might occur if this becomes public, but the primary harm described is to the affected communities, not to the organization deploying the system.

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- **Why D is incorrect:** Environmental harm relates to "Carbon footprint of large AI models, resource drains (lithium batteries, water)." The scenario describes social harm, not environmental resource consumption.

Question 8

What is the key difference between transparency and explainability in AI systems?

A. They are identical concepts with no meaningful difference
B. Transparency provides information about AI systems, while explainability makes the reasoning behind specific decisions understandable
C. Transparency is only for technical teams, explainability is only for end users
D. Transparency is required by law, explainability is optional

Correct Answer: B

Explanation:

- **Why B is correct:** The training distinguishes these concepts: "Being transparent about automated processes is an important aspect of strong AI governance" involves disclosure of information about the system, while "Explainability is the capacity to describe an AI model and its expected impact and potential biases. It requires an understanding of how an AI model operates." Transparency means providing information about what the system does, how it was built, and when it's being used. Explainability means making individual decisions interpretable so stakeholders can understand why a specific outcome occurred. You can be transparent without being explainable (e.g., disclosing that you use a black-box neural network) and potentially explainable without full transparency (e.g., explaining decisions without revealing proprietary algorithms).
- **Why A is incorrect:** The training treats them as related but distinct concepts. Both are necessary for trustworthy AI, but they address different aspects—information disclosure vs. decision interpretability.
- **Why C is incorrect:** Both transparency and explainability should serve multiple stakeholders including technical teams, end users, regulators, and affected parties. The training doesn't limit either concept to specific audiences.
- **Why D is incorrect:** Both may be required by various laws and regulations (e.g., GDPR's right to explanation, various AI regulations requiring transparency). The training presents both as important responsible AI principles, not distinguishing their legal requirements.

Question 9

An organization using third-party AI vendors for customer service chatbots wants to demonstrate accountability. According to Module 2, what does accountability in AI governance primarily mean?

A. The vendor is fully responsible since they developed the AI
B. The organization is ultimately responsible for the AI they deploy, regardless of the number of contributors

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C. Responsibility is shared equally among all parties
D. No one is responsible if the AI was developed by a third party

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly states: "Accountable – Organizations ultimately need to be responsible for the AI they deliver, irrespective of the number of contributors... An organization should ensure that the AI systems they use are safe, secure, resilient, valid, reliable and fair." This is a fundamental principle: deploying organizations cannot outsource accountability. Even when using third-party AI, the deploying organization remains responsible for ensuring the system meets ethical, legal, and safety standards.
- **Why A is incorrect:** While vendors have responsibilities, accountability principles establish that the deploying organization cannot simply defer responsibility to vendors. The organization chose to deploy the system and is accountable for its impacts.
- **Why C is incorrect:** While multiple parties may have various responsibilities, the training emphasizes that the deploying organization is "ultimately" responsible. Accountability doesn't diffuse equally—the organization using the AI bears primary accountability for its deployment and impacts.
- **Why D is incorrect:** This contradicts the fundamental accountability principle. The training makes clear that organizations cannot avoid responsibility by using third-party technology. They must still ensure the AI meets appropriate standards.

Question 10

A loan approval AI system denies applications at different rates for different demographic groups, even when creditworthiness indicators are similar. The system is proprietary, and the vendor refuses to provide detailed explanations of how decisions are made. Which TWO responsible AI principles are MOST directly in tension here?

A. Fairness and transparency/explainability
B. Privacy and security
C. Safety and accountability
D. Inclusiveness and efficiency

Correct Answer: A

Explanation:

- **Why A is correct:** This scenario directly involves two principles: (1) Fairness—"protection from unfair bias" is violated by differential denial rates for similar creditworthiness across demographics, (2) Transparency/explainability—the proprietary black-box nature and vendor's refusal to explain prevents understanding why decisions differ. The training emphasizes both "Fairness" and "Transparency and explainability" as core OECD principles and responsible AI requirements. Without transparency, it's difficult to diagnose and remediate the fairness problem. These principles are in tension

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because the vendor prioritizes proprietary protection over explainability needed to assess fairness.

- **Why B is incorrect:** Privacy and security aren't the primary issues described. The scenario involves discriminatory outcomes and lack of explanation, not data protection or security vulnerabilities.
- **Why C is incorrect:** Safety relates to physical harm and system reliability. Accountability is relevant (who's responsible?), but the core tension is between unfair outcomes and inability to explain them, not safety concerns.
- **Why D is incorrect:** While inclusiveness relates to fairness, and efficiency might explain vendor reluctance to provide explanations, these aren't the primary responsible AI principles at stake. The training specifically identifies fairness and transparency/explainability as core principles.

Question 11

An AI system for medical diagnosis was trained on data from patients aged 18-65. It now makes errors when diagnosing children and elderly patients. Which concept does this BEST illustrate?

A. Temporal bias
B. Edge cases and outliers
C. Sampling bias
D. Implicit bias

Correct Answer: B

Explanation:

- **Why B is correct:** The training defines edge cases and outliers as "data outside the training set boundaries; includes noise and outliers (data points outside normal distribution)." Children and elderly patients fall outside the age range (18-65) the model was trained on, making them edge cases. The model hasn't learned patterns for these populations and performs poorly when encountering data outside its training boundaries.
- **Why A is incorrect:** Temporal bias involves models becoming outdated over time as conditions change. The issue here isn't that the model aged poorly—it's that certain populations were never included in training data.
- **Why C is incorrect:** While this could be framed as sampling bias (the training sample didn't represent the full patient population), the training specifically discusses edge cases and outliers as a distinct concept describing data outside training boundaries. Given that children and elderly are explicitly outside the training range, edge cases is more precise.
- **Why D is incorrect:** Implicit bias refers to prejudice by developers. There's no indication of intentional discrimination—the issue is that the training data range excluded certain age groups, creating a technical limitation.

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Question 12

According to the training, Amazon's 2014-2017 AI recruiting system was abandoned because it was found to be biased against women. This is an example of which type of AI harm?

A. Societal harm affecting democratic processes
B. Individual harm in employment/hiring discrimination
C. Environmental harm from resource consumption
D. Organizational harm through cultural damage

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly lists this under "Individual harms – Bias and discrimination" with examples including "Employment/hiring: Amazon, 2014: implemented an AI system to help with recruiting and hiring; during testing they found the system was biased against women... project was eventually abandoned in 2017." Employment discrimination is an individual harm affecting people's economic opportunities and civil rights.
- **Why A is incorrect:** Societal harms in the training refer to "Spread of disinformation," "Ideological bubbles, echo chambers," "Deepfakes (elections, media manipulation)," and "Lethal autonomous weapons." While widespread employment discrimination has societal implications, the training categorizes this specific example under individual harms.
- **Why C is incorrect:** Environmental harm relates to "Carbon footprint of large AI models, resource drains (lithium batteries, water)." The Amazon recruiting case involves discrimination, not environmental impact.
- **Why D is incorrect:** While Amazon may have experienced organizational harm (reputational damage), the training presents this case as an example of individual harm to women who were discriminated against in the hiring process.

Question 13

A governance team is implementing strategies for trustworthy AI. According to the training, what is a recommended approach for reviewing higher-risk AI use cases that create ethical gray areas?

A. Assign responsibility to a single AI ethics officer
B. Develop a cross-functional and demographically-diverse oversight body
C. Outsource all ethical reviews to external consultants
D. Let the technical development team make all ethical decisions

Correct Answer: B

Explanation:

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- **Why B is correct:** The training explicitly recommends: "Develop a cross-functional and demographically-diverse oversight body to review higher-risk AI use cases that create ethical gray areas for the organization." Cross-functional ensures multiple perspectives (legal, technical, business, ethics), while demographic diversity brings different lived experiences that help identify potential harms to various groups. This aligns with the comprehensive governance approach the training advocates.
- **Why A is incorrect:** While having dedicated ethics roles is valuable, relying on a single person is insufficient for complex ethical questions. The training recommends a diverse body, not individual decision-making, to ensure multiple perspectives.
- **Why C is incorrect:** External consultants can provide expertise, but the training recommends establishing internal or external governance review boards. Organizations should develop internal capacity for ethical review, possibly supplemented by external input, not completely outsourced.
- **Why D is incorrect:** Technical teams have important expertise but lack the diverse perspectives needed for comprehensive ethical review. The training emphasizes cross-functional collaboration, not isolating ethical decisions within technical teams.

Question 14

An AI system trained on historical loan data from the 1990s and 2000s is used in 2024. The model performs well on traditional borrowers but poorly on applicants with modern financial profiles (cryptocurrency holdings, gig economy income, digital payment histories). This BEST illustrates which combination of issues?

A. Temporal bias and edge cases
B. Only sampling bias
C. Only implicit bias
D. Overfitting exclusively

Correct Answer: A

Explanation:

- **Why A is correct:** This scenario involves both concepts: (1) Temporal bias—"A model is trained and functions properly at the time, but may not work well at a future point" because financial landscapes changed dramatically from the 1990s-2000s to 2024 (cryptocurrency, gig economy, digital finance didn't exist or were minimal in training data), (2) Edge cases—modern financial profiles fall "outside the training set boundaries" since the model never learned about these new financial instruments and income sources. These borrowers represent outliers relative to the historical training distribution.
- **Why B is incorrect:** While sampling is involved, the key issue is that the data is outdated (temporal) and new financial types are outside training boundaries (edge cases), not just that the original sample was skewed toward a subset.

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- **Why C is incorrect:** There's no indication of developer prejudice. The problem is technical—outdated training data and new financial profiles the model never encountered, not implicit bias from developers.
- **Why D is incorrect:** Overfitting means the model is too fitted to training data patterns and doesn't generalize. While there might be some overfitting, the primary issues are that the data is old (temporal) and modern finance represents new cases outside the training range (edge cases).

Question 15

According to FIPs (Fair Information Practices), which principle requires that organizations implement security measures to protect personal data against risks such as loss, unauthorized access, destruction, or disclosure?

A. Notice/openness
B. Safeguards/security
C. Accountability
D. Access/individual participation

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly lists "Safeguards/security" as one of the FIP principles, which addresses implementing security measures to protect data. This principle requires organizations to use reasonable security safeguards to protect personal information against risks like loss, unauthorized access, destruction, use, modification, or disclosure.
- **Why A is incorrect:** "Notice/openness" requires organizations to be transparent about their data practices and inform individuals about data collection and use, not to implement security measures.
- **Why C is incorrect:** "Accountability" requires organizations to be responsible for complying with FIPs and often involves having someone responsible for compliance, but it's not specifically about security measures protecting data.
- **Why D is incorrect:** "Access/individual participation" gives individuals the right to access their data and request corrections. While important, it's not about implementing security safeguards.

Question 16

An organization discovers that its AI system for résumé screening performs well on most applications but makes errors on résumés with non-traditional formats, non-English names, or career gaps. According to the training, what does this scenario BEST represent?

A. The system is working perfectly and needs no changes
B. Edge cases and outliers that the system wasn't adequately trained to handle

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C. Temporal bias from outdated training data
D. Purely technical issue with no governance implications

Correct Answer: B

Explanation:

- **Why B is correct:** These résumés represent "Edge cases and outliers: data outside the training set boundaries." Non-traditional formats, non-English names, and career gaps likely appeared less frequently or not at all in training data, making them outliers. The training notes edge cases "includes noise and outliers (data points outside normal distribution)" and that they can cause errors. This has significant governance implications: the system may disadvantage non-traditional candidates, international applicants, and those with career interruptions (often women, caregivers, or people with disabilities).
- **Why A is incorrect:** Errors on certain types of résumés indicate a fairness and performance problem that needs addressing. The training emphasizes protecting against unfair bias and ensuring AI works fairly across diverse populations.
- **Why C is incorrect:** Temporal bias involves data becoming outdated over time. The issue here isn't that résumé formats changed—it's that certain formats and names were underrepresented in training data from the start.
- **Why D is incorrect:** This has significant governance implications. Errors on non-English names could constitute national origin discrimination; errors on career gaps might disproportionately affect women and people with disabilities. The training emphasizes addressing bias throughout the AI lifecycle.

Question 17

According to the OECD AI Principles, which principle specifically addresses the need for AI systems to function appropriately, be secure against attacks, and pose no unacceptable safety risks?

A. Human rights and democratic values
B. Transparency and explainability
C. Robustness, security, and safety
D. Accountability

Correct Answer: C

Explanation:

- **Why C is correct:** The training lists OECD AI Principle #4 as "Robustness, security, and safety," which directly addresses these technical requirements. This principle ensures AI systems are reliable, resilient to attacks, and don't create unacceptable safety risks—covering both cybersecurity and physical safety concerns.
- **Why A is incorrect:** "Human rights and democratic values" (Principle #2) addresses fairness, privacy, and respect for fundamental rights, not technical robustness, security, and safety requirements.

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- **Why B is incorrect:** "Transparency and explainability" (Principle #3) focuses on making AI understandable and providing information about how it works, not on technical security and safety requirements.
- **Why D is incorrect:** "Accountability" (Principle #5) addresses organizational responsibility for AI systems' actions and impacts, not the technical requirements for robustness, security, and safety.

Question 18

A content moderation AI system works well in English but fails to detect harmful content in other languages, particularly those using non-Latin scripts. This causes harm to users in those language communities. What type(s) of bias and harm does this scenario PRIMARILY involve?

A. Sampling bias leading to group harms
B. Only temporal bias with no real harm
C. Implicit bias affecting individuals only
D. No bias present, just technical limitations

Correct Answer: A

Explanation:

- **Why A is correct:** This involves sampling bias because "Data gets skewed toward a subset of a group and therefore may favor that subset of a larger group"—the training data was skewed toward English content. This creates group harms because entire language communities (often correlating with ethnic/national groups) receive inadequate protection from harmful content. The training identifies "Group harms" including discrimination and inadequate service to subgroups. This scenario demonstrates both technical bias and real-world harm to communities.
- **Why B is incorrect:** Temporal bias involves data becoming outdated over time. The issue here is underrepresentation of non-English languages from the start, not data aging. There is real harm—users in non-English communities are exposed to harmful content that isn't moderated.
- **Why C is incorrect:** While individuals in non-English communities are harmed, this is primarily a group-level harm affecting entire language communities systematically, not isolated individual impacts. Implicit bias involves developer prejudice, which may or may not be present—the sampling bias is the clearer issue.
- **Why D is incorrect:** Technical limitations that result in discriminatory performance constitute bias from a governance perspective. The training emphasizes that AI systems must work fairly across diverse populations, and differential performance based on language/ethnicity is a fairness concern, not an acceptable "technical limitation."

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According to the training, which of the following is a recommended practice for creating a culture of ethical AI within an organization?

A. Focus only on technical team training
B. Train and educate employees across all relevant functions including legal compliance, equitable design, transparency, privacy, and data governance
C. Ethical AI culture will develop naturally without intervention
D. Only executive leadership needs ethics training

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly states: "Train and educate employees to create a culture of ethical AI," including "legal compliance, equitable design, transparency and explainability, privacy and cybersecurity, data governance." Comprehensive education across functions ensures everyone understands their role in responsible AI, from developers to legal teams to business stakeholders. The training also emphasizes "provide knowledge resources and training to personnel" as part of creating a responsible AI culture.
- **Why A is incorrect:** While technical teams need training, ethical AI requires cross-functional engagement. Legal teams need to understand compliance, business teams need to understand risks, and all employees should understand ethical principles relevant to their roles.
- **Why C is incorrect:** The training explicitly recommends proactive culture-building through training, education, policies, and dedicated efforts. Ethical culture requires intentional development, not passive hope it will emerge naturally.
- **Why D is incorrect:** While leadership commitment is crucial, ethical AI culture requires organization-wide understanding and engagement. Limiting training to executives leaves most employees unprepared to identify and address ethical issues in their daily work.

Question 20

An environmental advocacy group raises concerns that training large language models produces carbon emissions equivalent to several cars' lifetimes and consumes substantial water resources. According to Module 2, what type of harm does this represent?

A. Individual harm affecting economic opportunity
B. Group harm affecting specific communities
C. Environmental harm from AI resource consumption
D. Organizational harm through reputational damage

Correct Answer: C

Explanation:

- **Why C is correct:** The training explicitly identifies "Environmental harms: Carbon footprint of large AI models, resource drains (lithium batteries, water)" as a distinct

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category of AI harms. It notes "AI training large models emits CO2 equivalent to several cars' lifetimes, high water/energy use." This scenario directly describes environmental harm from AI's resource consumption and carbon footprint.

- **Why A is incorrect:** Individual economic harms involve issues like "job displacement, discriminatory AI-driven hiring, market/job access bias." The scenario describes environmental impact, not individual economic opportunity.
- **Why B is incorrect:** While environmental harm might disproportionately affect certain communities, the training categorizes environmental impact as its own distinct harm type, separate from group harms which focus on discrimination and social divides.
- **Why D is incorrect:** While the organization might face reputational harm if environmental concerns become public, the primary harm described is the actual environmental impact—carbon emissions and resource consumption—not organizational reputation.

Question 21

A financial services company wants to use differential privacy to protect customer data in its AI training process. What does differential privacy primarily accomplish?

A. Completely eliminates all privacy risks
B. Adds statistical noise to protect individual data points while preserving aggregate patterns
C. Removes all personal data from datasets
D. Encrypts data so it can never be decrypted

Correct Answer: B

Explanation:

- **Why B is correct:** Differential privacy is a PET that "adds noise to protect individual data points" while maintaining the statistical properties of the dataset as a whole. It ensures that removing or adding a single individual's data doesn't significantly change the output, protecting individual privacy while allowing aggregate analysis. This provides mathematical privacy guarantees.
- **Why A is incorrect:** No technology completely eliminates all privacy risks. Differential privacy provides strong protections with mathematical guarantees, but the training doesn't claim any PET eliminates all risks. Implementing differential privacy requires careful parameter selection, and excessive noise can reduce model utility.
- **Why C is incorrect:** This describes de-identification, not differential privacy. Differential privacy works with actual data but adds mathematical noise to protect individuals. De-identification removes identifying information.
- **Why D is incorrect:** This describes encryption, not differential privacy. Homomorphic encryption enables computation on encrypted data, but differential privacy uses a different approach—adding calibrated noise to outputs or intermediate computations.

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Question 22

An AI system for determining social benefit eligibility replicates patterns from historical data where certain economic groups were systematically denied benefits. What responsible AI principle is MOST directly violated, and what is the primary concern?

A. Transparency; users don't know how decisions are made
B. Fairness; the system perpetuates historical discrimination against economically disadvantaged groups
C. Privacy; personal financial data is being processed
D. Safety; the system might malfunction

Correct Answer: B

Explanation:

- **Why B is correct:** The training specifically lists "Insurance/social benefits: If the system is not appropriately modeled and developed, there can be a discriminatory impact against particular groups of individuals, often based on economic status." Replicating historical discrimination patterns violates fairness principles and creates "Individual harms – Bias and discrimination." The primary concern is that the AI perpetuates and potentially amplifies past inequities, systematically disadvantaging economically vulnerable populations. The training emphasizes "protection from unfair bias" as a core responsible AI requirement.
- **Why A is incorrect:** While transparency is important and lack of it can compound the problem, the primary violation is fairness—the system produces discriminatory outcomes. Even a transparent system that discriminates would violate fairness principles.
- **Why C is incorrect:** Processing financial data engages privacy considerations, but the primary issue described is discrimination based on economic status, not privacy violation. Privacy concerns would focus on unauthorized access or excessive collection.
- **Why D is incorrect:** Safety typically refers to physical safety or technical reliability preventing harm. The issue here is systematic discrimination, which is a fairness concern, not a safety malfunction. The system is "working" as trained—the problem is that the training data embeds discrimination.

Question 23

According to the training, what is a potential organizational harm from deploying AI systems too rapidly without adequate testing and governance?

A. Improved efficiency eliminates all risks
B. Acceleration risks including deploying systems before adequate risk assessment and controls are in place
C. Faster deployment always benefits the organization
D. Speed has no relationship to organizational harm

Correct Answer: B

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Explanation:

- **Why B is correct:** The training explicitly identifies "Organizational harms: Reputational, cultural, economic, acceleration, legal/regulatory" with "acceleration" representing the risk of moving too fast. This aligns with the earlier discussion of "Speed and scale: Rapid deployment risks: The speed at which AI technologies can be developed and deployed outpaces regulatory responses, raising concerns about unforeseen consequences." Accelerated deployment without adequate governance can lead to failures, regulatory violations, discrimination, or other harms that damage the organization.
- **Why A is incorrect:** Efficiency is a potential benefit, but it doesn't eliminate risks. The training extensively discusses various risks that faster deployment can exacerbate, including insufficient testing, inadequate bias assessment, and compliance failures.
- **Why C is incorrect:** This ignores the "acceleration" category of organizational harm. Faster deployment can harm organizations if systems are deployed before proper risk assessment, testing, and controls are in place, leading to failures or regulatory issues.
- **Why D is incorrect:** The training explicitly includes "acceleration" as a type of organizational harm, demonstrating that deployment speed directly relates to organizational risk. The training also discusses rapid deployment risks throughout Module 1 and 2.

Question 24

A company is evaluating whether to use homomorphic encryption for sensitive data in its AI system. What is the PRIMARY advantage of homomorphic encryption?

A. It's faster than processing unencrypted data
B. It enables computation on encrypted data without decryption
C. It removes all identifying information from data
D. It prevents any unauthorized access permanently

Correct Answer: B

Explanation:

- **Why B is correct:** The training describes homomorphic encryption as one of the PETs that "enables computation on encrypted data." This is homomorphic encryption's defining feature: data remains encrypted while being processed, and results are obtained in encrypted form that can be decrypted by authorized parties. This means sensitive data never needs to be exposed in plain text during computation, providing strong privacy protection.
- **Why A is incorrect:** Homomorphic encryption typically adds significant computational overhead compared to processing unencrypted data. The privacy benefits come at the cost of slower processing, not faster.
- **Why C is incorrect:** This describes de-identification or anonymization, not homomorphic encryption. Homomorphic encryption keeps data encrypted during processing; it doesn't remove identifying information.

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- **Why D is incorrect:** While homomorphic encryption provides strong protection during processing, no technology prevents all unauthorized access permanently. It's a powerful privacy tool but not an absolute guarantee. Other security layers (access controls, encryption at rest, etc.) are also needed.

Question 25

According to the training, deepfakes that manipulate political figures' images and voices to spread false information during elections represent which category of AI harm?

A. Individual harm to the political figure only
B. Organizational harm to media companies
C. Societal harm affecting democratic processes
D. Environmental harm from computing resources

Correct Answer: C

Explanation:

- **Why C is correct:** The training explicitly lists "Societal harms: Spread of disinformation," "Ideological bubbles, echo chambers," and "Deepfakes (elections, media manipulation)" as examples of societal-level harms. Deepfakes in elections threaten democratic processes by enabling sophisticated disinformation, undermining public trust, and potentially influencing electoral outcomes through false information. This affects society broadly, not just individuals.
- **Why A is incorrect:** While the depicted political figure may experience individual harm (reputational damage), the primary concern identified in the training is the broader societal impact on democratic processes and public discourse, not just individual harm to one person.
- **Why B is incorrect:** Media organizations might be affected, but the training categorizes election deepfakes as societal harm because they threaten democratic integrity and public trust, not primarily as organizational harm to media companies.
- **Why D is incorrect:** While creating deepfakes consumes computing resources, the training discusses election deepfakes under societal harms related to disinformation and democratic processes, not under environmental harms from resource consumption.

Question 26

An AI system achieves optimal mathematical accuracy by fitting closely to training data patterns, but this results in poor performance on new data. This BEST illustrates which type of bias?

A. Sampling bias
B. Temporal bias
C. Overfitting to training data
D. Implicit bias

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Correct Answer: C

Explanation:

- **Why C is correct:** The training defines overfitting as: "A model works for the training data, but does not work for new data because it is so fitted to the training data." The scenario explicitly describes a model that performs well on training data (optimal mathematical accuracy) but poorly on new data—the textbook definition of overfitting. The model learned noise and specific patterns in the training data rather than generalizable patterns.
- **Why A is incorrect:** Sampling bias involves training data skewed toward a subset of the population. While sampling bias can contribute to overfitting, the scenario specifically describes the overfitting phenomenon—excellent training performance but poor generalization—not just biased sampling.
- **Why B is incorrect:** Temporal bias occurs when data becomes outdated over time. The issue here is overfitting to training patterns, not that the data has aged or conditions have changed temporally.
- **Why D is incorrect:** Implicit bias refers to prejudice by developers toward particular groups. Overfitting is a technical machine learning problem about model complexity and generalization, not developer prejudice.

Question 27

According to FIPs, the principle that requires organizations to keep personal data accurate, complete, and up-to-date is:

A. Data quality/relevance
B. Purpose specification
C. Safeguards/security
D. Use limitation

Correct Answer: A

Explanation:

- **Why A is correct:** The training lists "Data quality/relevance" as one of the FIP principles. This principle requires that personal data should be relevant to the purposes for which it will be used and should be accurate, complete, and kept up-to-date. Poor data quality can lead to incorrect decisions, particularly harmful in AI systems that make automated decisions.
- **Why B is incorrect:** "Purpose specification" requires that purposes for data collection be specified at or before collection time, not that data be kept accurate and current.
- **Why C is incorrect:** "Safeguards/security" addresses protecting data from unauthorized access, loss, or misuse through security measures, not data accuracy and currency.

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- **Why D is incorrect:** "Use limitation" requires that data only be used for specified purposes and not for other purposes without consent, not that data be accurate and current.

Question 28

A tenant screening AI system uses historical rental data that includes landlords' past decisions, some of which were discriminatory. The AI learns to replicate these discriminatory patterns. This scenario BEST illustrates which governance challenge?

A. AI systems can perpetuate and amplify historical biases embedded in training data
B. AI automatically eliminates human bias
C. Historical data is always neutral and unbiased
D. Discrimination only occurs in new systems, not from historical data

Correct Answer: A

Explanation:

- **Why A is correct:** This is a fundamental AI governance challenge the training emphasizes throughout: AI systems trained on historical data can learn and perpetuate historical discrimination. The training specifically mentions "Housing: Tenant selection and mortgage qualification can be affected if a biased AI system is used" under individual harms. The training emphasizes "Bias and discrimination" risks and notes that "Data quality and bias: Poor-quality or biased data can lead to flawed AI outputs." Historical data often reflects past discrimination, and AI systems trained on this data can automate and scale those biases.
- **Why B is incorrect:** This is the opposite of what the training teaches. AI systems can amplify human biases present in training data rather than eliminating them. The training extensively discusses how AI can perpetuate discrimination.
- **Why C is incorrect:** Historical data frequently reflects past discrimination, inequities, and biases. The training's extensive discussion of sampling bias, discrimination harms, and the need for data quality assessment directly contradicts the idea that historical data is inherently neutral.
- **Why D is incorrect:** The training makes clear that discrimination often stems from historical patterns in training data. Many examples of AI discrimination (Amazon recruiting, facial recognition, etc.) involve learning from biased historical data, not just new system design flaws.

Question 29

According to the training, which strategy is recommended for addressing privacy concerns in AI systems?

A. Collect as much data as possible to improve model accuracy
B. Implement data minimization, ensuring personal data unlikely to improve the model is excluded by default

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C. Privacy protections aren't necessary for AI systems
D. Use all available personal data without restriction

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly recommends: "Data minimization: while it takes a massive volume of data to develop and train AI models, personal data that is unlikely to improve the model should be left out by default." This aligns with FIPs' "Data minimization/collection limitation" principle and addresses privacy concerns by limiting exposure of personal data to what's necessary. The training also emphasizes using PETs and privacy-protective measures.
- **Why A is incorrect:** This contradicts data minimization principles. While AI benefits from data, the training emphasizes collecting and using only what's necessary, not maximizing data collection. "Data dependency" is listed as a governance challenge requiring careful data management.
- **Why C is incorrect:** The training extensively discusses privacy as a core concern, listing it in the OECD AI Principles, FIPs, and responsible AI principles. Privacy is identified as crucial for trustworthy AI.
- **Why D is incorrect:** This violates multiple principles including data minimization, use limitation, and purpose specification. The training emphasizes restricting data use to what's necessary and appropriate, with proper protections.

Question 30

An organization's AI system makes decisions that significantly affect individuals' access to education. According to the training, what governance approach is MOST appropriate given the high-stakes nature of these decisions?

A. Minimal governance since AI is more accurate than humans
B. Rigorous governance including fairness testing, transparency, explainability, human oversight, and appeal mechanisms
C. No human involvement to ensure objectivity
D. Governance is unnecessary for education decisions

Correct Answer: B

Explanation:

- **Why B is correct:** The training identifies "Education: AI systems used to select individuals to attend a school. A biased system can discriminate against qualified individuals based on race, gender or economic background" as a significant individual harm. High-stakes decisions affecting education access require comprehensive governance: fairness testing to prevent discrimination, transparency about the system's use, explainability so decisions can be understood, human oversight to catch errors, and appeal mechanisms for affected individuals. The training's emphasis on "Human rights and democratic values" and "appropriate choices about use of personal data and human intervention in decision-making" supports this approach. The OECD principle of

accountability requires ensuring systems are "safe, secure, resilient, valid, reliable and fair."

- **Why A is incorrect:** High-stakes decisions require MORE governance, not less, regardless of AI accuracy claims. The training emphasizes comprehensive governance especially for systems affecting civil rights and opportunities. Accuracy alone doesn't ensure fairness or protect rights.
- **Why C is incorrect:** The training emphasizes "human-centricity" and "appropriate... human intervention in decision-making" as responsible AI principles. High-stakes decisions should involve human oversight, not eliminate it for supposed objectivity. Complete automation of such decisions poses accountability and fairness risks.
- **Why D is incorrect:** Education decisions significantly affect individuals' opportunities and potential civil rights. The training's extensive discussion of harms, bias, and responsible AI principles makes clear that high-stakes AI applications require robust governance.

End of Module 2 Advanced Practice Test

Scoring Guide:

- **27-30 correct:** Exceptional understanding - exam-ready
- **24-26 correct:** Strong understanding - minor review needed
- **21-23 correct:** Good foundation - focus on application scenarios
- **18-20 correct:** Adequate knowledge - more practice with bias types and principles needed
- **Below 18:** Review Module 2 thoroughly, especially OECD principles, FIPs, and harm taxonomies

Key Concepts to Master for Module 2:

1. **Types of Bias:**
    - Implicit, sampling, temporal, overfitting, edge cases/outliers
2. **OECD AI Principles (5 principles):**
    - Inclusive growth, human rights/democratic values, transparency/explainability, robustness/security/safety, accountability
3. **Fair Information Practices (FIPs):**
    - Data minimization, use limitation, safeguards, notice, access, accountability, purpose specification, data quality
4. **Categories of AI Harms:**
    - Individual, group, societal, organizational, environmental
5. **Privacy-Enhancing Technologies:**

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    - Differential privacy, federated learning, homomorphic encryption, de-identification

6. **Responsible AI Principles:**
    - Fairness, transparency, explainability, accountability, privacy, security, safety, human-centricity
7. **Key Distinctions:**
    - Transparency vs. explainability
    - Different types of bias and when they apply
    - Different harm categories and examples

Study Tips:

- Focus on applying concepts to scenarios, not just memorizing definitions
- Understand WHY each principle matters and what problems it addresses
- Practice identifying which type of bias or harm applies to different situations
- Know the relationship between principles and frameworks
- Understand how different PETs work and when to use each

Good luck with your AIGP certification!

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